

BETHEL, AK, USA

WMO: 702190

Lat: 60.785N

Lon: 161.829W

Elev: 102

StdP: 14.64

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 26615

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.8	-20.8	-29.9	1.0	-23.9	-25.8	1.3	-19.7	35.9	13.4	33.0	7.8	9.2	20	0.833

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.3	73.1	59.2	69.4	57.5	65.9	56.1	61.0	69.1	59.2	66.4	57.5	63.8	10.2	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
58.2	72.7	62.9	56.2	67.6	60.8	54.4	63.1	59.1	27.1	68.8	25.8	66.6	24.8	63.8	72.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 29.1	26.1	23.5	-28.7	79.8	5.9	5.4	-32.9	83.7	-36.4	86.8	-39.7	89.9	-44.0	93.8		
(5)			-27.1	63.7	4.5	3.3	-30.3	66.1	-33.0	68.0	-35.4	69.9	-38.7	72.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	31.6	7.5	14.2	14.4	28.8	42.8	53.2	56.0	53.8	46.3	32.1	18.4	10.1		
	DBStd	20.92	17.58	15.43	13.70	10.95	7.75	5.44	5.12	4.68	5.75	9.13	12.78	15.69		
	HDD50	7221	1317	1001	1104	637	242	25	7	13	133	555	949	1238		
	HDD65	12211	1782	1421	1569	1087	687	354	282	348	560	1019	1399	1703		
	CDD50	490	0	0	0	0	20	123	193	131	23	0	0	0		
	CDD65	7	0	0	0	0	0	2	4	1	0	0	0	0		
	CDH74	101	0	0	0	0	1	34	52	14	0	0	0	0		
	CDH80	20	0	0	0	0	0	4	14	2	0	0	0	0		
	WSAvg	11.3	13.8	13.4	12.5	11.5	10.4	9.5	9.5	10.2	10.1	11.0	11.9	12.5		
	PrecAvg	19.3	0.8	0.7	0.7	0.7	1.1	1.7	2.6	3.1	2.9	1.7	1.8	1.2		
(15) Precipitation	PrecMax	23.8	2.6	3.0	3.5	2.2	2.6	4.2	4.1	6.5	6.9	4.4	4.2	3.5		
	PrecMin	11.8	0.1	0.1	0.1	0.1	0.2	0.4	0.7	1.0	0.8	0.2	0.1	0.3		
	PrecStd	3.1	0.7	0.7	0.7	0.6	0.7	0.7	1.0	1.2	1.5	1.0	1.0	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.5	39.9	42.3	55.8	70.5	78.7	80.2	75.3	64.6	53.6	44.5	40.9		
		MCWB	36.8	36.2	37.6	44.8	53.5	59.7	63.5	61.8	54.1	47.4	41.5	36.8		
	2%	DB	37.5	37.4	37.8	50.9	64.7	72.7	73.3	69.0	60.1	49.8	40.2	37.4		
		MCWB	34.7	34.2	34.6	42.3	51.5	57.6	60.3	59.0	52.7	45.8	38.3	34.9		
	5%	DB	35.3	35.9	35.8	46.7	59.8	68.3	69.6	65.2	57.2	46.7	37.4	35.2		
		MCWB	33.4	33.6	33.5	40.4	49.0	55.7	58.5	57.1	51.5	43.9	36.0	33.5		
	10%	DB	32.5	34.0	33.7	42.9	56.0	64.4	65.7	62.0	54.5	44.2	35.0	32.7		
		MCWB	31.4	32.5	32.1	38.1	46.9	54.1	57.1	56.1	50.4	42.1	33.9	31.5		
	0.4%	WB	37.2	36.8	38.1	45.9	55.2	62.0	66.6	64.1	56.4	49.3	42.0	37.4		
		MCDB	39.9	39.1	41.5	54.1	66.5	74.2	78.2	71.5	61.1	51.7	43.5	39.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	35.0	35.0	35.1	42.7	52.3	58.9	61.5	60.7	54.0	46.4	38.7	35.2		
		MCDB	37.1	36.6	37.2	49.9	62.6	69.3	70.1	66.4	58.1	49.0	39.9	36.9		
	5%	WB	33.5	33.8	33.5	40.6	50.1	57.0	59.5	58.7	52.3	44.4	36.1	33.6		
		MCDB	35.1	35.4	35.2	46.4	59.1	66.7	67.7	63.2	55.8	46.6	37.3	35.0		
	10%	WB	31.5	32.5	32.0	38.3	47.7	55.0	57.9	56.9	50.9	42.2	33.9	31.6		
		MCDB	32.8	34.1	33.8	42.6	55.3	63.0	64.5	61.1	54.0	44.1	34.8	33.0		
	5% DB	MDBR	10.7	12.5	14.1	14.9	16.4	16.2	13.3	11.9	12.2	10.7	10.6	11.5		
		MCDBR	11.6	11.4	13.2	19.7	23.7	23.9	21.3	17.9	15.8	12.1	9.4	10.8		
		MCWBR	10.3	9.7	10.7	12.6	12.4	10.7	9.4	8.8	9.2	8.9	8.4	9.6		
		MCWBR	11.1	11.0	12.4	18.5	22.8	21.3	18.9	15.2	13.2	10.9	9.3	11.0		
(40) Clear-Sky Solar Irradiance	taub		0.227	0.285	0.289	0.354	0.376	0.365	0.373	0.381	0.341	0.295	0.248	0.215		
		taud	2.086	2.156	2.228	2.145	2.293	2.405	2.418	2.382	2.455	2.420	2.073	1.927		
	Ebn at Noon		167	220	265	264	268	274	268	255	247	222	148	106		
		Edh at Noon	13	23	29	39	36	33	32	31	24	17	13	9		
(44) All-Sky Solar Radiation	RadAvg		118	341	924	1424	1606	1650	1359	1066	751	387	138	59		
	RadStd		18	43	97	122	150	142	134	122	85	29	20	8		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (7 neighbors)		N/A	+3.01	+2.88	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page